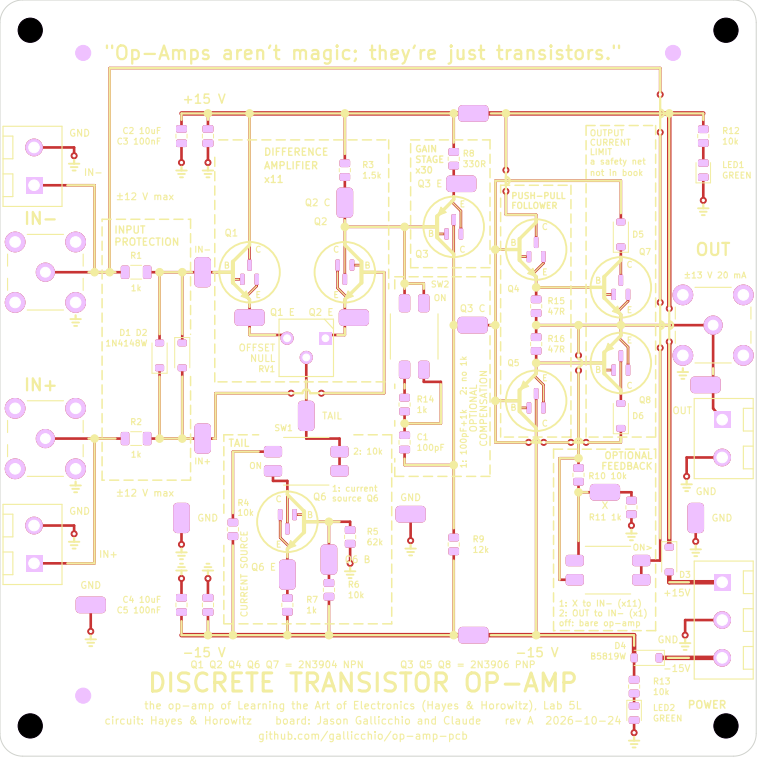

"Op-Amps aren't magic; they're just transistors."
+15 V
GND
C2 10uF
C3 100nF
IN-
±12 V max
INPUT PROTECTION
R1 1k
D1 D2 1N4148W
R2 1k
IN+
C4 10uF
C5 100nF
DIFFERENCE AMPLIFIER x11
R3 1.5k
Q1
Q2
Q1 E
Q2 E
OFFSET NULL RV1
TAIL
SW1
2: 10k
CURRENT SOURCE
Q6
1: current source Q6
R4 10k
R5 62k
R6 10k
R7 1k
Q6 E
GAIN STAGE x30
Q3 E
Q3 C
SW2
R14 1k
C1 100pF
1: 100pF c1 2: no 1k
OPTIONAL COMPENSATION
R8 330R
R9 12k
PUSH-PULL FOLLOWER
Q4
Q5
R15 47R
R16 47R
D5
D6
Q7
Q8
OUTPUT CURRENT LIMIT
a safety net not in book
OPTIONAL FEEDBACK
R10 10k
R11 1k
1: X to IN- (x1)
2: OUT to IN- (x1)
off: bare op-amp
R12 10k
LED1 GREEN
OUT
±13 V 20 mA
−15 V
Q1 Q2 Q4 Q6 Q7 = 2N3904 NPN
Q3 Q5 Q8 = 2N3906 PNP
Q4 B5819W
R13 10k
LED2 GREEN
POWER
DISCRETE TRANSISTOR OP-AMP
the op-amp of Learning the Art of Electronics (Hayes & Horowitz), Lab 5L
circuit: Hayes & Horowitz board: Jason Gallicchio and Claude rev A 2026-10-24
github.com/gallicchio/op-amp-pcb